

Problem

If the sawtooth waveform in Fig. 17.9 (see Practice Prob. 17.2) is the voltage source $v_s(t)$ in the circuit of Fig. 17.22, find the response $v_o(t)$.

Figure 17.22:

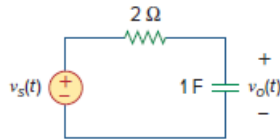
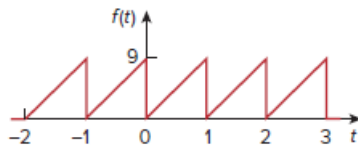


Figure 17.9:



Step-by-step solution

Step 1 of 2

Refer to Figure 17.9 in the textbook for the sawtooth waveform.

The Fourier series of the sawtooth waveform is,

$$v_s(t) = 4.5 - \frac{9}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin 2\pi n t$$

Consider that this sawtooth waveform is applied to the circuit in Figure 17.22 in the textbook.

Apply voltage division method on the circuit to find the response.

$$\begin{aligned} \mathbf{V}_o &= \frac{\frac{1}{j\omega_n C}}{R + \frac{1}{j\omega_n C}} \mathbf{V}_s \\ &= \frac{\frac{1}{j\omega_n C}}{\frac{j\omega_n RC + 1}{j\omega_n C}} \mathbf{V}_s \\ &= \frac{\mathbf{V}_s}{1 + j\omega_n RC} \end{aligned}$$

Here, the angular frequency from the input waveform, $v_s(t)$ is,

$$\begin{aligned} \omega_n &= n\omega_0 \\ &= 2\pi n \text{ rad/s} \end{aligned}$$

Substitute 2Ω for R , 1 F for C , and $2\pi n$ for ω_n .

$$\begin{aligned} v_o(t) &= \frac{\mathbf{V}_s}{1 + j(2\pi n)(2 \Omega)(1 \text{ F})} \\ &= \frac{\mathbf{V}_s}{1 + j4\pi n} \end{aligned}$$

Step 2 of 2

Consider that $n = 0$ for the dc component. Therefore,

$$\mathbf{V}_s = 4.5$$

Find the response of the circuit.

$$\begin{aligned}\mathbf{V}_o &= \frac{4.5}{1+j(0)} \\ &= 4.5 \text{ V}\end{aligned}$$

The ac component of the input waveform for n th harmonic is,

$$\mathbf{V}_s = \frac{9}{n\pi} \angle 90^\circ$$

Find the response of the circuit.

$$\begin{aligned}\mathbf{V}_o &= \left(\frac{1}{1+j4\pi n} \right) \left(\frac{9}{n\pi} \angle 90^\circ \right) \\ &= \left(\frac{1}{\sqrt{1+16\pi^2 n^2} \angle \tan^{-1}(4\pi n)} \right) \left(\frac{9}{n\pi} \angle 90^\circ \right)\end{aligned}$$

Therefore, the complete response in time domain is,

$$\begin{aligned}v_o(t) &= \frac{9}{2} - \sum_{n=1}^{\infty} \frac{9}{n\pi\sqrt{1+16\pi^2 n^2}} \cos[n\omega_0 t + 90^\circ - \tan^{-1}(4\pi n)] \\ &= \frac{9}{2} - \frac{9}{\pi} \sum_{n=1}^{\infty} \frac{\sin[n\omega_0 t - \tan^{-1}(4\pi n)]}{n\sqrt{1+16\pi^2 n^2}} \quad (\text{since, } \cos(90^\circ - \theta) = \sin \theta) \\ &= \frac{9}{2} - \frac{9}{\pi} \sum_{n=1}^{\infty} \frac{\sin[2\pi n t - \tan^{-1}(4\pi n)]}{n\sqrt{1+16\pi^2 n^2}} \quad (\text{since, } \omega_0 = 2\pi n)\end{aligned}$$

Thus, the response of the circuit is,

$$\boxed{\frac{9}{2} - \frac{9}{\pi} \sum_{n=1}^{\infty} \frac{\sin[2\pi n t - \angle \tan^{-1}(4\pi n)]}{n\sqrt{1+16\pi^2 n^2}} \text{ V}}$$